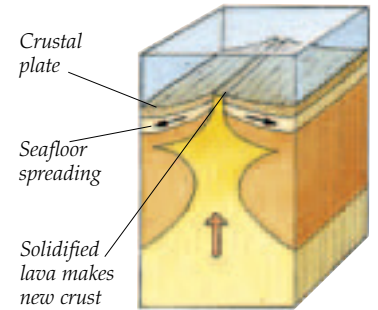




Vents and smokers

IN PARTS OF THE OCEAN FLOOR, there are cracks from which very hot, mineral-rich water gushes. These vents, or hot springs, exist at the spreading centers where gigantic plates that make up the Earth's crust are moving apart. Cold seawater sinks deep into cracks in the crust where the water is heated, collecting dissolved minerals. At temperatures of up to 750°F (400°C), hot water spews out and some of the minerals form chimneys (black smokers). Hot water produced by vents helps bacteria to grow, which create food from the hydrogen sulfide in the water. Extraordinary animals crowd around the cracks and rely on these microbes for food. In the late 1970s, scientists using submersibles discovered the first vent communities in the Pacific. Since then, vents have been discovered in other spreading centers in the Pacific and the Mid-Atlantic Ridge.



GROWING OCEAN

New areas of ocean floor are continually being created at spreading centers between two crustal plates. When hot, molten rock (lava) emerges from within the crust, the lava cools and solidifies adding material to the edge of each adjoining plate. Old areas of ocean floor are destroyed where one plate slides under another. Lava from volcanic eruptions at spreading centers can kill off communities of vent animals.

Animals cook if too close to hot water in a vent



BLACK SMOKER

Animal life abounds in an active vent site, such as this one in the Mid-Atlantic Ridge. If the vent stops producing hot, sulfur-rich water, the community is doomed. Animals from dying vents must colonize a new site, which may be several hundred miles away across the cold, almost barren seafloor.



Plumes of hot water are rich in sulfides, which are poisonous to most animals

Dense numbers of animals crowd around a vent

Fish predators nibble tops off tube worms

Giant clams in the eastern Pacific can grow to 12 in (30 cm) long

Some animals graze on mats of bacteria covering rocks near a vent

Model of hydrothermal vents found in the eastern Pacific

